

Case Study:
How Algorithm Oppressed Women In Hiring Process

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Abstract

This paper examines the impact of algorithmic bias on gender equity in hiring practices, using Amazon's AI-powered recruiting tool as a case study. Despite the increasing use of algorithms in recruitment to save time and costs, the inner workings of these systems can be difficult to observe, and their behavior is largely determined by the data that is fed into them. In Amazon's case, the algorithm was trained on a biased dataset consisting mostly of male resumes, resulting in gender bias that favored male candidates for technical positions. This paper argues that companies must exercise careful evaluation of the algorithms they use in their hiring processes, ensuring impartial and transparent design and application. Furthermore, it suggests that organizations should prioritize accountability and openness in their recruiting procedures, allowing for frequent audits and reviews to guarantee impartial and fair results. Overall, this paper highlights the importance of understanding the ethical implications of AI and machine learning in hiring practices and taking proactive measures to reduce any potential biases and ensure equitable hiring procedures.

Background

Gender equity is a broad and hot issue; however, to narrow the gap we need to focus on specific aspect of this topic. This research focuses on one factor - hiring process - which affected gender pay gap. The gender pay gap is defined as the disparity in earnings between men and women. Meanwhile, an analysis of the labor market reveals that the participation rate for women is 76.9% while for men it is 88.5%, showing a gap of 11.6%. Similarly, the employment rate for women is 74.7% compared to 85.8% for men, leading to a gap of 11.1%(U.S. Department of

labor, 2023). In STEM fields, women are still not proportionally represented and they earn lower salaries than their male counterparts. According to the U.S. Census Bureau, in 2019, women made up only 27% of STEM workers, which is an increase from 8% in 1970.

An algorithm refers to a systematic method consisting of a set of rules and steps that aid in resolving a problem or accomplishing a task. Nowadays, many organizations use algorithms to evaluate job candidates based on diverse metrics such as qualifications, experience, and expertise. Nevertheless, the use of algorithms in hiring practices has led to concerns regarding the possible emergence of biases and discrimination. Hence, it is crucial for employers to exercise careful evaluation of the algorithms they employ in the hiring process and make certain that they are impartial and transparent in their design and application.

Case: Amazon Recruit System (2018)

Large companies attract many applicants because of their variety of positions and higher salaries than their peers, but the huge number of applications has also led such large companies to use machine learning algorithms to screen resumes to reduce labor costs. However, after introducing algorithm to the hiring process, does it fair to gender equity? In the case of Amazon's recruiting system, we see the bias.

Since 2014, Amazon started building programs with the aim of mechanizing the search for top talent. According to sources familiar with the matter, the organization utilized an experimental AI-powered hiring tool that assigned job applicants scores on a scale of one to five stars, akin to how products are rated by consumers on Amazon's platform. In 2015, the company

discovered that their new system was not evaluating candidates for technical positions, such as software developers, in a gender-neutral manner.

To understand the topic at hand, it's important to provide some context about AI algorithms. These complex systems are often described as black boxes, meaning that the inner workings of the algorithm are not easily observable. Nevertheless, the algorithm's behavior is largely determined by the data that is fed into it. For example, consider training an algorithm to recognize faces in a race. If we provide the algorithm with more pictures of white faces than black faces, it may have a higher accuracy in identifying white individuals while exhibiting bias when attempting to differentiate black faces.

Consider the data that Amazon utilized to train its hiring model - the company fed in all of the resumes they received over the course of ten years. However, since men have historically held more prominent positions in the tech industry, it follows that a majority of these resumes were from male candidates. In essence, the algorithm was trained on a biased dataset, and this can be likened to a facial recognition algorithm that was trained on more images of white faces than black faces. Therefore, it's not unreasonable to assume that Amazon's hiring model would exhibit gender bias and be less accurate when evaluating resumes from female candidates.

There's nothing wrong with using the resumes your company has received over the past decade -- it's the most accessible source of data. The problem is that at the beginning of designing this algorithm, the staff did not design it from the perspective of data ethics. Of course, professional technicians will not be ignorant of the impact of training data on model output, but at the beginning of the design, people ignore the consideration of gender biased by default, and take it for granted by default.

To investigate the sources of gender bias in recruiting algorithms, a 2018 Reuters report found that these technologies exhibit a preference for candidates whose resumes contain verbs commonly used on male engineers' resumes, such as "executed" and "captured." While the underlying reasons for these differences between male and female resumes remain unclear, the evidence suggests that the algorithm favors the most commonly occurring patterns in men's portfolios. Moreover, the algorithm penalizes the ranking of resumes containing keywords such as "women," as seen with the example of "women's chess club captain."

These findings emphasize the risk that hiring algorithms will reinforce gender biases, especially in industries where women are underrepresented. The removal of talented female candidates due to the usage of language associated with fields with a male predominance as well as the punishment of resumes that contain phrases associated with femininity might further restrict chances for women. To support fair and equitable hiring procedures, it is essential to identify and mitigate these biases in recruiting algorithms. Future studies should look into further reasons why hiring algorithms are biased towards women and devise methods to get rid of them.

Suggested Solutions

To ensure that algorithms used in the hiring process are inclusive of a diverse candidate pool, it is essential for organizations to thoughtfully select and balance the data used in training these algorithms. Moreover, it is imperative to involve individuals from various backgrounds in the development and testing of algorithms, to detect and eliminate any potential biases.

Companies should prioritize transparency and accountability in their recruitment processes,

permitting frequent audits and evaluations to ensure impartial and just outcomes. Furthermore, third-party auditing and certification can provide an additional layer of oversight, keeping businesses accountable for their recruitment procedures. Ultimately, it is vital for companies to recognize the ethical implications of utilizing AI and machine learning in the hiring process, and to take proactive measures to mitigate any biases, creating a fair and equitable hiring experience for all.

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